

EM-PPCA Sentiment Index

VADER Inputs · 1995 Q1 – 2025 Q4 · Quarterly

Model Configuration

Parameter	Value
Method	Probabilistic PCA (PPCA) via EM
Factors	1
Centre / Scale	Yes / Yes
Max iter / Threshold	1,000 / 10^{-5}
Output	Z-scored ($\mu = 0$, $\sigma = 1$)
Seed	42

Factor Results — Sentiment Index - VADER

Variable	Loading	λ	Var.%	R^2	Missing
gdelt_avgTone	−0.641				
fed_minutes_vader	+0.599				
gov_vader	+0.051	1.670	33.40%	0.344	3.09%
fed_speech_vader	−0.474				
sec_vader	+0.052				

Red = negative loading Green = positive loading Sign flipped post-estimation.

Factor Results — PC1 (FinBERT)

Variable	Loading	λ	Var.%	R^2	Missing
gdelt_avgTone	−0.608				
fed_minutes_finbert	+0.338				
gov_finbert	+0.310	1.731	34.63%	0.356	3.09%
fed_speech_finbert	−0.441				
sec_finbert	+0.475				

Red = negative loading Green = positive loading Sign flipped post-estimation.